

## **Battery Cost Savings Model**

### **1. Purpose of the Model**

This model assesses the cost-effectiveness of installing a battery storage system in a solar-powered household. It quantifies savings in grid electricity costs when excess solar power is stored and reused, comparing scenarios with and without a battery over a 20-year horizon, under two electricity price inflation assumptions.

### **2. Data and Checks**

Data Source: solar\_energy\_data.csv - hourly electricity usage and solar generation for 2020. 2020 is a leap year, but no data was provided for 29 February, so the file holds 365 days of hourly records.

Columns:

- Hour
- time\_stamp
- generation\_(kWh)
- usage\_(kWh)

Checks performed:

- Valid timestamp format
- 8760 rows (= 365 days x 24 hours); no missing values
- Outliers investigated: (a) a usage reading of 46,000 kWh - clearly erroneous, replaced with the mean of the adjacent hours; (b) five negative usage readings - impossible for household usage, treated as sign errors and corrected using the absolute value; (c) a single generation reading of 13.05 kWh at 01:00 - physically impossible night-time solar output, replaced with the mean of the adjacent hours. The recurring evening usage spikes (>25 kWh) were investigated and retained as legitimate high demand. No cap was applied to generation: 12.5 kWh is the battery capacity, not the panel output.
- Hourly average plots confirmed expected residential patterns after correcting outliers

- After these corrections the data was deemed complete and fit for analysis.

### **3. Assumptions Used**

- Battery cost: \$7,000
- Lifetime: 20 years
- Max capacity: 12.5 kWh
- Discount rate: 6% per annum
- Electricity price (2022): \$0.17/kWh
- Scenario 1: 4% inflation annually (Government estimate)
- Scenario 2: Starts at 4%, increases by 0.25% yearly (Naomi's estimate)
- Usage and generation patterns remain constant
- No battery degradation or feed-in tariffs considered.

### **4. Methodology**

#### Hourly Modeling:

- Calculate grid electricity needed (no battery):  $\max(\text{usage} - \text{generation}, 0)$
- Calculate excess solar:  $\max(\text{generation} - \text{usage}, 0)$
- Simulate battery storage: store excess, discharge when needed
- Cap battery charge at 12.5 kWh, minimum 0 kWh

#### Monthly Aggregates:

- Summarized total generation, usage, and grid electricity for each calendar month

#### Financial Projection (20 years):

- Base savings (kWh) from 2020 scaled by forecast annual electricity prices; year 1 (2022) is priced at \$0.17/kWh and the first escalation is applied on 1 Jan 2023
- Calculate NPV at 6% discount rate
- Calculate IRR using root-finding techniques

Each scenario (with and without battery, and under both inflation forecasts) is modeled separately.

## **5. Financial Results Summary**

Extra electricity met by the panel + battery combination compared with panels alone: 3,499.85 kWh/year; implied saving at 1 Jan 2022 prices: \$594.97 per year.

Scenario 1 (Government estimate - 4% inflation):

- NPV of savings: approximately \$ 9,424.32 (net benefit after the \$7,000 cost: \$2,424.32)
- IRR: approximately 9.43%
- Conclusion: Financially viable (gross PV of savings comfortably exceeds the \$7,000 cost)

Scenario 2 (Naomi estimate - increasing inflation):

- NPV of savings: approximately \$ 10,734.40 (net benefit after the \$7,000 cost: \$3,734.40)
- IRR: approximately 10.71%
- Conclusion: Financially viable investment

## **6. Reasonableness and Validation Checks**

- Battery charge correctly bounded between 0 and 12.5 kWh.
- Hourly flows and balances checked via plots.
- Monthly aggregates match hourly totals.
- IRR calculations consistent with NPV trend.

Results are consistent with expectations for a typical solar-powered household with battery storage.

## **7. Recommendations**

- Battery installation is financially justifiable under both scenarios (net benefit is positive at the 6% discount rate in each case).
- Assuming energy prices rise at 4% per year or more, the battery system yields between 9.43% and 10.71% annual returns — making it a strong long-term investment.

Suggestions:

- Investigate potential feed-in tariff policies.
- Consider lower battery installation costs.
- Explore higher usage scenarios or time-of-use tariffs.

Model is reusable for other households and future projections.